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        ROLL NO - 49  
        PRACTICAL NO - 02
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In [3]: # importing libraries  
import numpy as np  
  
# function for threshold gate  
def linear_threshold_gate(dot, T):  
    '''Returns the binary threshold output'''  
    if dot >= T:  
        return 1  
    else:  
        return 0  
  
# matrix of inputs  
input_table = np.array([  
    [0, 0], # both no  
    [0, 1], # one no, one yes  
    [1, 0], # one yes, one no  
    [1, 1] # both yes  
])  
  
print(f'Input table:\n{input_table}')  
  
# weights  
weights = np.array([1, -1])  
  
# dot product  
dot_products = input_table @ weights  
  
# threshold  
T = 1  
  
# loop through inputs  
for i in range(4):  
    activation = linear_threshold_gate(dot_products[i], T)  
    print(f'Input: {input_table[i]}, Dot Product: {dot_products[i]}, Activation: {activation}')
```

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Input table:  
[[0 0]  
 [0 1]  
 [1 0]  
 [1 1]]  
Input: [0 0], Dot Product: 0, Activation: 0  
Input: [0 1], Dot Product: -1, Activation: 0  
Input: [1 0], Dot Product: 1, Activation: 1  
Input: [1 1], Dot Product: 0, Activation: 0
```

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In [ ]:
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